PROJECT DOCUMENTATION

Student Performance Analytics System

Python with AI • Mini Project

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GitHub Repository	https://github.com/Mrrudra55

Submitted as part of the Python with AI project assignment

1. Project Title and Student Details

Project Title: Student Performance Analytics System

Student Name: Rudra Pratap Nayak

Course: BCA

Institution: N.C. Autonomous College, Jajpur

GitHub Repository: https://github.com/Mrrudra55

2. Objective

The objective of this project is to build a Python-based system that reads student records from a CSV dataset, processes the data, and presents useful information about academic performance. The system calculates total and average marks, assigns grades, identifies pass/fail status, compares subject performance, and highlights top-performing students. It also includes attendance and department-level analysis to make the results easier to explore.

The project demonstrates the use of Python fundamentals, functions, conditions, loops, NumPy, and Pandas for a practical data-analysis task. A Streamlit interface is also included to make the analysis accessible through a dashboard.

3. Technologies Used

Technology	Purpose
Python	Main programming language used to implement the project logic.
Pandas	Used to load the CSV dataset, organize records in a DataFrame, filter records, and perform analysis.
NumPy	Used for numerical operations and conditional classification such as pass/fail and support status.
Streamlit	Used to create an interactive dashboard for exploring the student data and analysis results.
CSV	Used as the data-file format for the student dataset and exported analysis report.
GitHub	Used to store and share the project source files and documentation.

4. Dataset Description

The project uses a CSV file named students.csv containing 20 sample student records. The dataset includes student identification details, department, attendance percentage, and marks in three subjects. The sample data is intended for demonstrating analysis and should not be treated as official academic records.

Column	Description
Student_ID	Unique identifier for each student.
Name	Student name.
Department	Department associated with the student record.
Attendance	Attendance percentage.
Python	Marks obtained in Python.
Java	Marks obtained in Java.
AI	Marks obtained in Artificial Intelligence.

5. Implementation

1. **Prepare the dataset:** Store student details, attendance, and subject marks in students.csv.
2. **Load the data:** Use Pandas to read the CSV file into a DataFrame.
3. **Calculate totals and averages:** Calculate each student's total marks across Python, Java, and AI, then calculate the average.
4. **Assign grades:** Assign a grade based on the student's average using the grade thresholds defined in the program.
5. **Identify pass/fail status:** Mark a student as Pass only when the student meets the minimum passing mark in every listed subject; otherwise mark Fail.
6. **Analyze subjects and class performance:** Calculate subject averages and class-level statistics, and identify highest- and lowest-performing results.
7. **Find top performers:** Sort the processed records by performance and display the top five students.
8. **Analyze attendance and support needs:** Classify attendance using a 75% threshold and identify records that may need academic support based on low attendance and/or failing marks.
9. **Analyze departments:** Summarize performance by department and count subject-wise pass/fail results.
10. **Export results:** Save the processed student analysis to student_analysis_report.csv.
11. **Dashboard:** Use app.py to provide a Streamlit dashboard with dataset upload, filters, search, charts/summary views, student exploration, and CSV downloads.

5.1 Project File Structure

```
Student_Performance_Analytics/
├── main.py
├── functions.py
├── app.py
├── students.csv
├── requirements.txt
├── README.md
└── Project_Documentation.docx
```

6. Key Features

- Loads student records from a CSV file using Pandas.
- Calculates total and average marks for Python, Java, and AI.
- Assigns grades using clearly defined average-mark thresholds.
- Determines pass/fail status based on the marks in each subject.
- Displays class statistics and subject-wise averages.
- Identifies the top five students.
- Analyzes attendance using a configurable minimum attendance threshold.
- Flags students who may need additional academic support.
- Provides department-level and subject-wise pass/fail analysis.
- Exports the processed report to a CSV file.
- Offers a Streamlit dashboard with data upload, search, filters, and downloadable results.

6.1 Grading Rule

Average Marks	Grade
90 and above	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
Below 50	F

Pass/fail rule: a student passes only when marks are at least 40 in every analyzed subject. Attendance rule: attendance of 75% or higher is classified as meeting the attendance threshold.

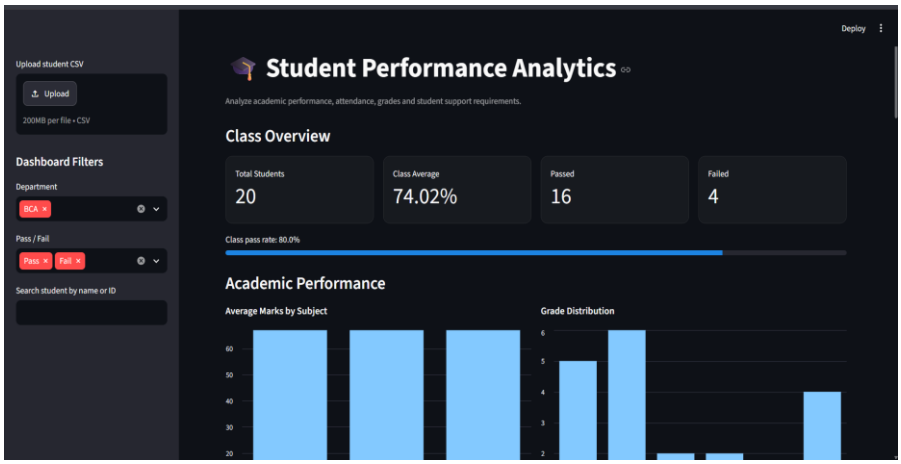
7. Output and Screenshots

The command-line program displays the processed student performance, class statistics, subject averages, top students, attendance/support information, department analysis, and subject-wise pass/fail counts. It also exports a CSV report. The Streamlit dashboard provides a visual way to inspect and filter the dataset.

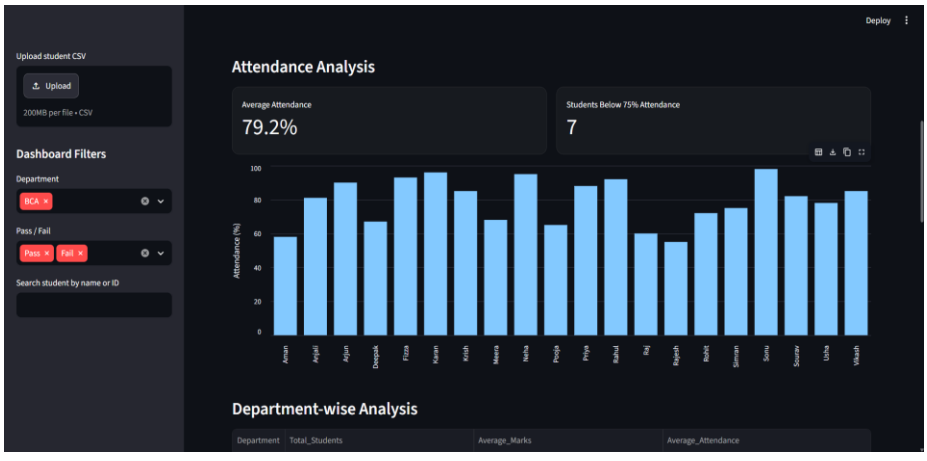
Add your own screenshots in the spaces below before final submission. Recommended screenshots:

- Screenshot 1: Terminal output after running python main.py.
- Screenshot 2: Streamlit dashboard showing the main metrics and charts.
- Screenshot 3: Student table with search or department/status filters.
- Screenshot 4: Downloaded/exported student_analysis_report.csv (optional).

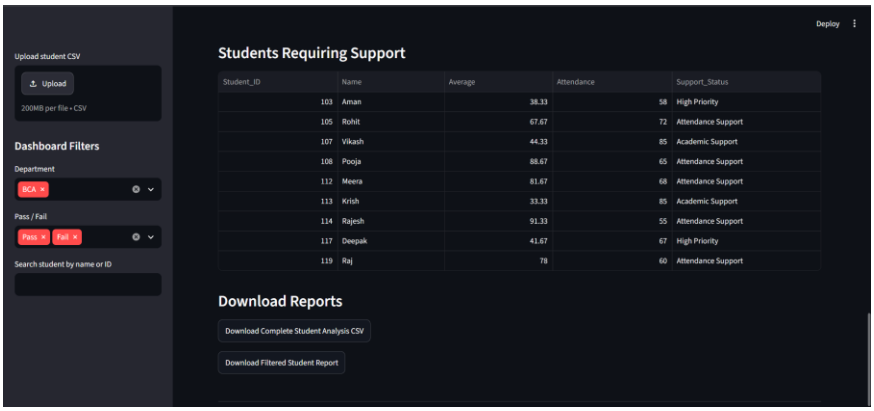
Screenshot 1 — Main program output



Screenshot 2 — Dashboard overview



Screenshot 3 — Student explorer



8. Final Outcome

The project provides a working workflow for loading student data, calculating performance metrics, categorizing results, and producing summary analysis. The reusable functions keep calculations organized, while the dashboard makes it easier to search and explore student records. The generated CSV report can be used for further review. The project demonstrates how Python, NumPy, and Pandas can be applied to a small real-world-style data-analysis problem.

9. Challenges and Learning

Challenges addressed during development included:

- Organizing the project into separate files for data, reusable functions, the main program, and the dashboard.
- Making sure numeric columns are suitable for calculations and analysis.
- Defining consistent rules for grades, pass/fail status, and attendance classification.
- Combining multiple analyses into a clear report and interactive dashboard.
- Checking that the program runs and that generated results can be exported.

Learning outcomes:

- Practiced reading and processing CSV data with Pandas.
- Used NumPy for numerical and conditional operations.
- Practiced writing reusable functions and applying conditions to data.
- Learned to summarize data by subject and department.
- Gained introductory experience building a data dashboard with Streamlit and preparing a GitHub repository.

10. GitHub Repository

The complete source code, dataset, README file, requirements file, and this documentation should be available in the GitHub repository.

GitHub URL: https://github.com/Mrrudra55

Before submitting the Google Form, open the repository in a browser and verify that the files are visible and the README explains how to run the project. If the trainer has specified repository visibility, follow those instructions.

Appendix: How to Run the Project

1. Install Python, if it is not already installed.

2. Open a terminal in the project folder.

3. Install dependencies using:

```
python -m pip install -r requirements.txt
```

4. Run the command-line program:

```
python main.py
```

5. To run the dashboard, use:

```
python -m streamlit run app.py
```

Note: The exact package list should match the project's requirements.txt. If a command or file name differs in your final repository, update this section accordingly.